SPEAKER CHEAT SHEET · HANDOUT

Speaker Cheat Sheet (English) — Louis Renner Masterclass

Presenter: Clemens Kunert · Louis Renner GmbH (Meuselwitz Plant, Germany)
Scope: 33 Slides across 5 Chapters · Synchronized with all finalized technical terminology

LOUIS RENNER GMBH
Since 1882 · Werk Meuselwitz

Live Stage Navigation: `→` / `Space` = advance · `←` = back · Click any chapter card on **Slide 2** to jump directly · `▸` marks interactive elements on the slide.

Chapter 1: Introduction, Heritage & Meuselwitz Plant (Slides 01–04)

SLIDE 01 LOUIS RENNER (Title Slide)

- **Core Message:** *The Heart of the Piano Since 1882* — over 140 years of action and hammerhead precision, now united under one roof in Meuselwitz.
- **Opening Line:** *"Welcome everyone—today we will walk from raw Merino wool in Wurzen through our unified hammerhead production in Meuselwitz and the drop-in Yamaha Kit all the way to the physics and craft of concert voicing."*

SLIDE 02 Structure and Chapters at a Glance

- **5 Chapters:**
  1. **History & Meuselwitz (Slides 3–4):** From 1882 in Stuttgart to our unified Meuselwitz plant.
  2. **Wurzen Felt Production (Slides 5–10):** Filzfabrik Wurzen since 1783, wool blend, carding, 56% fulling density, unsanded felt sheets.
  3. **Hammerhead Production (Slides 11–18):** Felt storage, Weickert Special SEM, cutting & underfelt, **hammer mouldings** & hammer weight, kettle & side jaws, CNC & robotic calibration, riveting & boring.
  4. **Yamaha Kit C1–C3 (Slides 19–24):** Ready-to-mount C1–C3 set in 4–5 hours on site, India field proof & acoustic analysis (`H₁–H₁₂` up to `C₈`).
  5. **Voicing & Open Q&A (Slides 25–33):** `A₁` partial buildup, complex spring, clock model, crown & string plane, fine voicing, lacquer/ironing & open Q&A.

SLIDE 03 140+ Years of Tradition & Perfection (1882 – Today)

- `▸ Interaction:` Step through all **9 milestones** using `→` / `Space`:
  1. **1882 Stuttgart:** Founded by Louis Renner — specialized precision mechanics focused on accurately fitted piano actions.
  2. **1906 In-House Hammerheads:** Dedicated hammerhead manufacturing launched (tracing the arc from Cristofori's 1700 leather-covered parchment roll to Renner).
  3. **1911 Grand Actions & Electrification:** Building on Érard's repetition principle; Oscar Renner replaces line shafts with hundreds of individual electric motors.
  4. **1944 Destruction & Rebuilding:** Complete destruction in the July 1944 Stuttgart air raid; determined reconstruction by workforce and family.
  5. **1951–1955 Dr. Manfred Renner:** Joins in 1952, returns permanently in 1955 — drives global expansion as benchmark supplier.
  6. **1990 Fall of the Wall:** Acquisition of Piano Union's action plant in Zeitz to safeguard major customers (e.g., Schimmel > 40 pianos/day).
  7. **2011 Two Core Sites:** Modern **Meuselwitz plant (`6,800 m²`)** after relocating from Zeitz + **Gärtringen facility** (specialized woodworking and administration).
  8. **2019 Steinway & Sons:** Joins the Steinway family to preserve German action and hammerhead craftsmanship long-term (honoring Clemens von Arnim).
  9. **Today — Unified Meuselwitz Plant:** Full consolidation of all production in Meuselwitz including a state-of-the-art central extraction system; gratitude to the Gärtringen workforce.

SLIDE 04 Construction Documentary: Building the Unified Meuselwitz Plant

- `▸ Interaction:` **14 flipbook construction phases (`11.2025 – 09.2026`)** followed by an **interior drone flight** through live production.
- **Focus:** Structural gutting, new administration & reception wing, central dust extraction, streamlined machine flow under one roof.

Chapter 2: The Art of Felt Manufacturing – FFW Wurzen (Slides 05–10)

SLIDE 05 Filzfabrik Wurzen (FFW) – Birthplace of Piano Hammer Felt

- ▸ Interaction: **6 interactive tabs** + **Fiber Lens** on the right.
- **Key Facts:** Founded in **1783** · In **1903**, Ernst Weickert developed modern tapered piano hammer felt in Wurzen (`Weickert Special`).
- **Takeaway:** Unique regional synergy between FFW Wurzen (felt) and Renner Meuselwitz (hammerheads).

SLIDE 06 Pure Merino Wool & the Secret of Fiber Blending

- **Stage 01 (Raw Wool & Scouring):** Fine-crimped pure Merino wool (Australia, New Zealand, South Africa, South America); **gentle scouring** protects epicuticle scales and retains natural wool wax (lanolin).
- **Stage 02 (Blending):** Proprietary balance of **long tensile fibers** (for outer crown tension) and **fine filler wool** (for resilient inner support).

SLIDE 07 Carding: Parallel Fiber Alignment into Fine Fleece Webs

- **Process:** Historic roller carding machines comb millions of crimped Merino fibers parallel into gossamer webs (`≤ 2%` basis-weight tolerance).
- **Impact:** Establishes exact fiber orientation and prepares the **wedge taper** from heavy bass to slender treble.

SLIDE 08 The 4 Pillars of Fulling

- **4 Pillars: 1. Moist Steam · 2. Controlled Warmth (~45 °C) · 3. Mechanical Pressure & Oscillating Motion · 4. Master Experience**.
- **Physics:** Warmth and moisture open the keratin scales just enough to interlock in 3D — **100% mechanical, zero chemical binders**.

SLIDE 09 56% Fulling Density (0.56 g/cm³): The Natural Limit

- **Key Number:** `56%` **(** `0.56 g/cm³` **)** is the physical ceiling of purely mechanical fulling.
- **Why It Matters:** Under-fulled felt lacks internal tension; over-fulling beyond 56% fractures keratin fibers (**felt breakage**, dead rebound).

SLIDE 10 Final Compaction & Tapered 36-Layer Anatomy

- ▸ Interaction: Toggle **Unsanded (Renner/FFW)** vs. **Sanded Sheet**.
- **36-Layer Build:** Staggered **shortened core webs** inside form the wedge (thick bass on the right, thin treble on the left) — enclosed by `2 × 6` **continuous outer webs**.
- **Key Point: Leaving the felt sheets unsanded** keeps all outer tension layers unbroken — preserving 100% tensile strength and preventing tearing when pressed around the **hammer moulding**!

Chapter 3: Hammerhead Production in Meuselwitz (Slides 11–18)

SLIDE 11 Climate-Controlled Felt Storage & Assortment Diversity

- **Top-Felt Inventory: Over 30** tapered FFW & Weickert formulas · regulated humidity, pheromone monitoring, and seasoning relieve fulling stresses.
- **Underfelt Stock: 6 standard colors** (**Red & Wine Red**, **Orange**, **Yellow**, **Blue**, **Black**) plus custom customer dyeings.

SLIDE 12 "Weickert Special" – Exclusive to Renner

- ▸ Interaction: **SEM Zoom Viewer** (`1× Overview`, `10× Fiber Detail`, `22× Cuticle Zoom`, `Live Loupe`, `Felt Strip W 1512 K`).
- **Accredited STFI Lab Report P2024 0903 (M. Kermer):**
  - `56%` **max. fulling density** (`0.56 g/cm³`) — purely mechanical, zero chemical hardeners.
  - `0.36%` **residual wool grease (ISO 3074)** — `+38%` higher natural lanolin retention vs. `0.26%` reference sample.
  - `−65%` **less swelling** in 48h climate cycling (`4.81%` vs. `13.69%` thickness change) + intact epicuticle scales under SEM.

SLIDE 13 Felt Preparation: Top-Felt Cutting & Underfelt Profiling

- **01 · Top-Felt Cutting (** `≤ 1 mm Tolerance` **):** 6-axis robot cold-cuts seasoned sheets distortion-free without thermal glazing + inline batch coding.
- **02 · Underfelt Milling (** `± 0.1 mm Tolerance` **):** Contour-milled coupling bridge; feathered tails milled to `± 0.1 mm` ensure a seamlessly closed contour right into the high treble.

SLIDE 14 The Hammer Moulding: Tonewoods, Properties & Profiles

- ▸ Interaction: Right media toggle `Profiles` (`Form A, B, US, CSB`) ↔ `Timber Stock`.
- **The 4 Moulding Woods (key distinctions):**
  1. **Hornbeam (`Weißbuche`):** Heavy hardwood — **very demanding to season**, yet once properly dried, an **ideal, uniform moulding wood**.
  2. **Maple (`Ahorn`):** Heavy, often **more elastic** hardwood — though **irregular sugar deposits** increase the risk of variance.
  3. **American Walnut: Highest-grade wood** — very tight growth rings, long-fibered, and **very light**, making it **ideal for a brilliant treble**.
  4. **Kotibé (`Mahogany`):** Comparatively **heavy due to its high natural oil content** — used selectively for specific applications.
- **Profiles:** Symmetrical (`Form B · US · CSB` for Steinway, Bösendorfer, Bechstein, Yamaha, Kawai, Schimmel, Blüthner, Förster, Sauter) vs. asymmetrical (`Form A` with offset moulding shoulder).

SLIDE 15 Optimal Hammerhead Weight Relative to String Weight

- ▸ Interaction: Register tabs (`① Bass` , `② Break −1 g` , `③ Treble`), click keys `A₀–C₈` , weight simulator (`Too Light` / `Equilibrium` / `Too Heavy`) + `Test Strike` .
- **Unified Mass Scale (`0.5–220 g`):**
  1. **Bass (`Keys 1–23` , copper-wound `90–210 g`):** The hammerhead (`~15 → 13 g`) is **always comparatively light relative to the heavy bass string** — so bass hammer weight is **held high as long as possible**.
  2. **Break (`Key 23 → 24` , plain wire = `3 strings per note` , `~31 g` combined):** Deliberate `−1.0 g` **step** (`13.0 → 12.0 g`) onto the lighter 3-string plain choir.
  3. **Treble (`to C₈`):** Even across all 3 strings per note, choir mass drops below `3 g` down to `~0.6 g` (`~0.2 g` per wire) with `½ T < 0.5 ms` → smooth taper to `5.0 g` (e.g., lightweight American walnut moulding) to prevent self-damping.

SLIDE 16 Pressing & Gluing to 0.1 mm

- **Pressing Sequence (`0.1 mm` Tolerance):**
  1. First, the felt is pressed **downward into the kettle** (`Pressgesenk`).
  2. Next, the heated **side jaws** (`Gegenlagen`) close in from both sides and press the felt firmly onto the glue surface of the **hammer moulding**.
- **Result:** High outer tensile tautness over a compressed, resilient inner cushion (foundation for `pp` to `ff`).

SLIDE 17 CNC Milling & Robotic Calibration

- **3-Step Precision Workflow:**
  1. **Moulding Milling (`0.1 mm`):** CNC milling machines first profile the **hammer mouldings** to a `0.1 mm` tolerance.
  2. **Post-Gluing Trim Milling:** After pressing and gluing, the felt overhang is removed and cleanly finish-milled.
  3. **Robotic Calibration:** Subsequently, a **robotic cell** precisely calibrates each individual hammerhead.

SLIDE 18 Automated Riveting & Boring

- **3 Key Features:**
  1. **Precision Boring Angle:** Exact depth, diameter (e.g., `ø 5.4 mm`), and individual **boring angle** per note.
  2. **Brass Riveting:** Permanently locks the bonded felt tails against opening under heavy concert play.
  3. **Radiused Hammer Tail:** Cleanly contoured tail radii for reliable checking against the backcheck felt.

Chapter 4: The Louis Renner Yamaha Kit C1–C3 (Slides 19–24)

SLIDE 19 Louis Renner Yamaha Kit (C1–C3) – Seamless Drop-In

- ▸ Interaction: **5 Tabs** (`SYSTEM` · `TOUCH` · `DROP-IN` · `TONE` · `TIME SAVED`) + Macro Zoom on the right.
  1. **System:** 5 years of R&D, proprietary measurement & production system → full concert tone **with zero manual pre-voicing**.
  2. **Touch:** Calibrated **walnut moulding**, quarter-sawn **hornbeam shanks**, pre-tensioned knuckles → lower inertia, crisp repetition.
  3. **Drop-In:** Precision-bored hammerheads pre-glued onto shanks with flanges using glass-hard **natural hide glue** → **1-screw mounting** directly onto the original Yamaha rail (`0 h` boring labor).
  4. **Tone: Pressed from a single homogeneous felt strip** → seamless register transitions, articulated bass, bell-clear treble.
  5. **Time Saved:** `4–5 hours` **on site** instead of `3–7 workshop days` → zero transport risk, direct voicing to room acoustics & artist preference.

SLIDE 20 Field Proof India (1/2): Time-Lapse Installation in One Day

- **On-Site Proof:** Complete conversion in **one workday (`10:00 – 18:00`)** right at the venue — removal, 1-screw mounting onto the original rail, fine regulation (blow distance, let-off, checking), tuning & concert test.

SLIDE 21 Field Proof India (2/2): Pianist Raaga in Action

- **Artist Verdict:** Immediate shift from hard attack to warm, carrying European concert tone, blooming treble, effortless `pp–ff` shading, and noticeably faster repetition.

SLIDE 22 Acoustic Analysis (1/3): Sound Spectrum Original vs. Yamaha Kit

- ▸ **Interaction:** Click keyboard / slide 2-octave window (`Original 0:00–5:17` vs. `Yamaha Kit 5:18–11:42`).
- **Finding:** Clear overtone gain (`H₂–H₈`) across the singing and brilliance registers with clean fundamental focus.

SLIDE 23 Acoustic Analysis (2/3): Partial Distribution (`H₁–H₁₂`) & 24 Semitones

- ▸ **Interaction:** Register buttons (`Bass` to `Brilliance`) & mode (`Relative H₁` / `Δ Gain`).
- **Harmonic Architecture: Even partials (`H₂, H₄, H₆, H₈`)** build warmth & body; **odd partials (`H₃, H₅, H₇`)** deliver brilliance & singing core — consistent note by note.

SLIDE 24 Acoustic Analysis (3/3): 4 Tonal Gains & Overtone Retention up to C_8

- **4 Headline Metrics:**
  1. **`+2.53 dB` (`+79%`)** gain in the singing register (`0.8–2.5 kHz`, cantabile).
  2. **`+1.87 dB` (`+54%`)** treble brilliance boost (`2.5–6 kHz`).
  3. **`−3.28 dB`** reduction of muddy low-bass boom (`50–250 Hz` → cleaner bass chord separation).
  4. **`+0.93 dB/kHz`** flatter spectral roll-off across `pp–ff` right up to `C₈` (`4,186 Hz`).

Chapter 5: The Art of Concert Voicing & Finale (Slides 25–33)

SLIDE 25 Partial Structure on A_1 (55 Hz): Felt Hardness & Strike Point

- ▸ **Interaction:** Felt hardness slider + strike-point slider (`x₀ / L` on `L = 1,560 mm`) + audio playback.
- **2 Acoustic Laws:**
  1. **Hardness & Contact Time `τc`:** Softer/needled crown (`τc ≈ 3.8 ms`) attenuates upper partials; firmer crown (`τc ≈ 1.3 ms`) boosts high partials in dB.
  2. **Young-Helmholtz Law (`x₀ / L`):** Striking at **`1/8 L`** to **`1/7 L`** places dissonant partials (`H₇` natural seventh, `H₉` second) near a node while preserving the singing major third `H₅`.

SLIDE 26 The Hammerhead as a Complex Spring: Cristofori to Zone Balance

- ▸ **Interaction:** Switch `1. Cristofori (1700)` ↔ `2. Renner Spring` + four 3-step controls.
- **Physics:** Cristofori's hollow parchment roll used pure ring tension with zero dead mass. The Renner head acts as a **graded tension spring**: **`100%` compaction at the crown → `½` (`50%`) at the widest flank**, anchored to the **hammer moulding** by glue and rivet.
- **Diagnostic Rule:** Crown/transition too hard → **no `pp`** · Low elasticity or locked shoulder → `τc` prolonged, **no `ff`**!

SLIDE 27 Voicing Zones & Needling Effect: The Renner Clock Model

- ▸ **Interaction:** Click voicing zones directly on the cross-section.
- **Golden Rule:** Always follow the wrapped wool fiber course and needle **perpendicular to the fiber (`⊥ 90°`)** across `9:00–3:00` so needles part fibers cleanly rather than severing them!
- **Clock Zones:**
  - **Blue (`9–10 / 2–3 o'clock`):** `ff` wedge — deep strokes stopping short of the underfelt (opens fortissimo projection).
  - **Green (`10–11 / 1–2 o'clock`):** `f` wedge — `4–6 mm` deep (opens the cantabile singing core).
  - **Orange (`11–12 / 12–1 o'clock`):** `p` wedge — `2.5–4 mm` (smooths dynamic transition).
  - **Purple (crown rim):** `pp` wedge — **max. `2 mm`**, strictly **between** string grooves!
  - **Specialty Stitch Above Rivet → Steeply Toward Flank:** Few, deep stitches activate lower tension reserve, shorten `τc`, and boost upper partials (`H₇–H₁₂`).
  - **Red (`12:00` deep crown core): Strict Taboo** — severs the tension arch permanently!

SLIDE 28 Crown & String Plane in 3-String Choirs: Phase & Cancellation

- ▸ **Interaction:** Tilt/skew slider on `A₃` (`220 Hz` trichord) + unison audio check.
- **Workshop Rule:** Even microsecond arrival skews (`Δt`) from a tilted crown or uneven string height create **comb-filter cancellations** and metallic buzz — **mate all 3 strings to 100% simultaneous contact before touching a voicing needle!**

SLIDE 29 String Preparation: Hitch Pins, Bridge Pins & Agraffes

- **4 Mandatory Pre-Voicing Steps:**
  1. Clean and inspect hitch pins.
  2. Seat strings solidly at the bridge pins using a brass punch.
  3. Space and level the 3-string plane behind the agraffes.
  4. Verify damper seating and 100% pure unison hammer contact.

SLIDE 30 Intermediate, Fine & Una Corda Voicing

- **5 Finishing Steps:**
  1. Dedicated fine needle holder (BDK fine holder).
  2. **Intermediate voicing** of the shoulder zone on a firm support pad (chromatic evening).
  3. **Fine crown voicing** in the grand: **max.** `2 mm` strictly **between string grooves** (Chop-Stick).
  4. Light crown dressing with a `320–400 grit` sanding paddle.
  5. **Una corda voicing:** Depress soft pedal, check shifted groove alignment, and lightly chamfer the left crown edge.

SLIDE 31 Lacquer Dosing, Heat Ironing (max. 135 °C), Crown Finish & Weight

- ▸ **Interaction:** Click lacquer zones on cross-section + toggle playing-time compaction.
- **01 · Lacquer Dosing & Mass Tuning:**
  - **At the Rivet (below hammer moulding):** Reinforces the lower foundation & lifts tension while protecting the elastic impact arch.
  - **Laterally Above the Hammer Moulding:** Light side dosing boosts high partials (`H₇–H₁₂`), **but compacts further over playing time!** Excess lacquer glues the flank and locks up the spring.
  - **Directly on the Crown (`12:00`): Strictly avoid!** (Glassy, nasal hardness & permanent damage).
  - **Fullness & Projection via Mass:** Increase hammer weight from middle to bass (`+m`); reduce weight toward treble (`−m`).
- **02 · Heat Ironing:** Strictly **max.** `135 °C` (prevents keratin scorching & brittleness).
- **03 · Ultra-Fine Crown Finish:** Micro-abrasive down to `≤ 15 µm` or a **Braun Series 1** foil shaver (trims stray fuzz fibers with `0%` felt loss).

SLIDE 32 The Renner Academy & Technical Master Dialogue

- **Invitation:** Hands-on masterclasses (*By Masters for Masters* on action geometry, hammerhead fitting, Yamaha Kit & concert voicing), **factory tours at our unified Meuselwitz plant**, and direct technical support.

SLIDE 33 Let the Perfect Hammer Sound! (Finale & Open Q&A)

- **Closing Line:** *"Meuselwitz precision and finest Wurzen Merino felt — Let the Perfect Hammer Sound! Thank you very much, and let's open the floor for your questions and workshop experiences."*